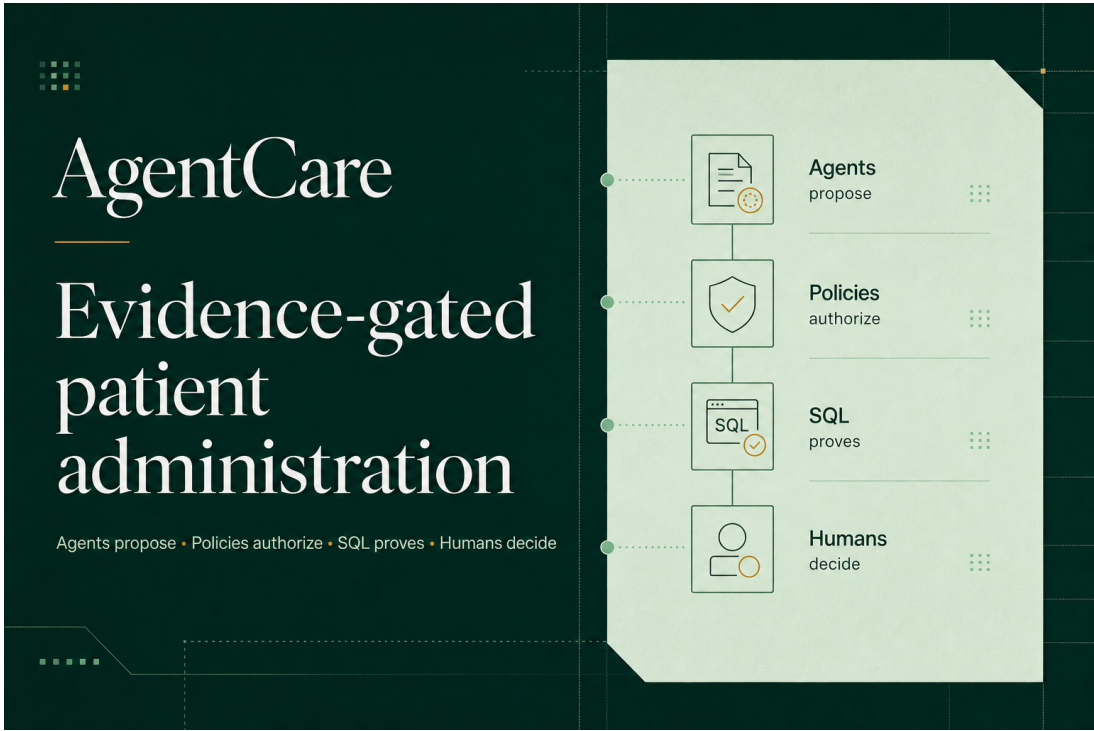




AgentCare Local Setup Guide

Clone, configure, run, verify, and test the complete evidence-gated patient administration platform.

Repository	github.com/chinmayk1613/AgentCare_AI_Patient_Administration_and_Care_Coordination
Branch	main
Recommended route	Docker Compose
Native stack	Python 3.11/3.12 + Node.js 22.13+ + pnpm 11.9
LLM	Optional developer-owned OpenAI key; safe fallback works without one

Agents propose. Policies authorize. Services execute. SQL proves. Humans decide exceptions.

Version 1.0 - July 28, 2026

1. What a new developer needs

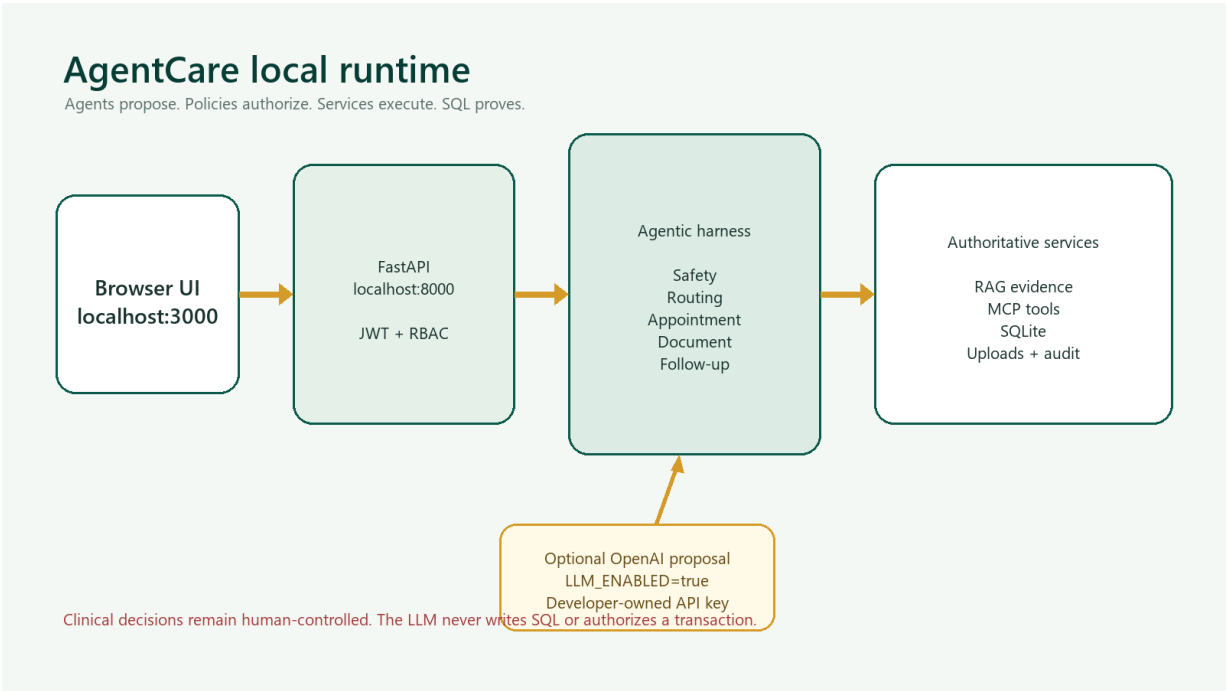
Choose the Docker route for the fastest complete startup. Choose the native route when editing or debugging the Python and TypeScript services separately.

Path	Required software	Best for
Docker - recommended	Git + Docker Desktop + Compose	Fast, isolated, complete startup
Native development	Git + Python 3.11/3.12 + Node 22.13+ + pnpm 11.9	Editing and debugging

- Optional OpenAI API key with active billing/quota for real LLM proposals.
- Editor such as VS Code.
- Synthetic document under 10 MB for upload tests.

No OpenAI key is required to exercise persistence, RAG retrieval, MCP tools, policy gates, booking, documents, human review, and audit evidence. The explicit safe-fallback path remains end-to-end.

2. Local architecture



The browser calls FastAPI locally. The backend owns JWT/RBAC, agents, RAG, MCP-style tools, SQLite transactions, uploads, and audit events. Only bounded proposals may use OpenAI.

3. Clone and inspect

Windows PowerShell, macOS, or Linux:

```
git clone https://github.com/chinmayk1613/AgentCare_AI_Patient_Administration_and_Care_Coordination.git
cd AgentCare_AI_Patient_Administration_and_Care_Coordination
git switch main
git status
```

Expected: branch main, clean working tree, and no real .env file or credential.

4. Recommended route - Docker Compose

Step 1 - Verify Docker Desktop is running

```
docker --version
docker compose version
```

Step 2 - Create the ignored local environment

```
Copy-Item .env.example .env      # Windows PowerShell
cp .env.example .env            # macOS/Linux

JWT_SECRET=replace-with-a-long-random-local-secret
LLM_ENABLED=false
OPENAI_API_KEY=
OPENAI_MODEL=gpt-5-mini
```

Step 3 - Build and start

```
docker compose up --build

- API: http://localhost:8000
- Web: http://localhost:3000
- API docs: http://localhost:8000/docs
- Persistent local volume: agentcare_data
```

Step 4 - Verify health

```
curl http://localhost:8000/health
# {"status":"ok","llm_mode":"safe-fallback","write_tools":true}
```

4. Docker lifecycle and reset

Stop while preserving the synthetic database and uploads:

```
docker compose down
```

Restart without rebuilding:

```
docker compose up
```

Delete only the local Docker volume and reseed:

```
docker compose down -v  
docker compose up --build
```

The -v command destroys the local AgentCare volume. Do not use it against important data.

5. Native backend setup

Step 1 - Verify runtimes

```
git --version  
python --version  
node --version  
pnpm --version
```

Expected: Python 3.11 or 3.12, Node.js 22.13+, and pnpm 11.9.

Step 2 - Create a virtual environment

```
# Windows PowerShell  
python -m venv .env  
./venv/Scripts/Activate.ps1  
python -m pip install --upgrade pip  
python -m pip install -r requirements.txt  
  
# macOS/Linux  
python3 -m venv .env  
source .env/bin/activate  
python -m pip install --upgrade pip  
python -m pip install -r requirements.txt
```

Step 3 - Copy backend environment template

```
Copy-Item backend\.env.example backend\.env # Windows  
cp backend/.env.example backend/.env       # macOS/Linux
```

5. Native services

Recommended backend/.env values:

```
APP_ENV=development
DATABASE_URL=sqlite:///./data/agentcare.db
JWT_SECRET=replace-with-a-long-random-local-secret
OPENAI_API_KEY=
OPENAI_MODEL=gpt-5-mini
LLM_ENABLED=false
UPLOAD_DIR=./data/uploads
MAX_UPLOAD_MB=10
WRITE_TOOLS_ENABLED=true
ALLOWED_ORIGINS=http://localhost:3000,http://localhost:5173
```

Step 4 - Seed and start FastAPI

```
cd backend
python -m app.seed
python -m uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Expected: backend/data/agentcare.db, synthetic hospital data, and HTTP 200 from /health.

Step 5 - Start the web interface in a second terminal

```
pnpm install --frozen-lockfile
pnpm run dev
```

Open <http://localhost:3000>. Localhost defaults to the API at <http://127.0.0.1:8000>.

6. Optional OpenAI LLM

```
LLM_ENABLED=true
OPENAI_API_KEY=your-own-key-here
OPENAI_MODEL=gpt-5-mini
```

Restart FastAPI. The /health response should show `llm_mode: enabled`. Each developer must use an authorized key with active quota.

Never commit or share an API key. A provider error returns the workflow to the safe-fallback path. The model never authorizes SQL writes or clinical decisions.

7. Synthetic accounts

Role	Email	Password
Patient	patient@agentcare.demo	Patient123!
Staff	staff@agentcare.demo	Staff123!
Reviewer	reviewer@agentcare.demo	Reviewer123!
Orthopedic reviewer	priya.orthopedics@agentcare.demo	Reviewer123!
Cardiology reviewer	elena.cardiology@agentcare.demo	Reviewer123!

Use synthetic data only. Additional demo accounts are listed in README.md.

8. End-to-end verification



A. Normal journey - Sign in as the patient and submit:

I need a cardiology follow-up next week. I also want to attach my previous ECG.

- Confirm stepwise safety, intent, RAG/MCP routing, live slot selection, appointment commit, document validation, reminder, case number, and audit evidence.

B. Human approval - Submit:

My legs are painful. I need to consult a doctor and submit my MRI report.

- Confirm staff review opens case details, approval resumes the same workflow, and live availability follows.

C. Document mismatch - Require MRI, upload a synthetic ECG, and verify MRI stays missing with a warning.

D. Safety boundary - Ask for diagnosis/prescription and verify escalation without clinical advice.

9. Automated tests

```
cd backend
python -m pytest
cd ..
```

```
pnpm run lint
pnpm run build
node --test tests/rendered-html.test.mjs
```

- Python source compiles and backend tests pass.
- Persistence, idempotency, safety, and RBAC behavior pass.
- TypeScript lint and Cloudflare Worker build pass.
- Rendered interface test confirms the real AgentCare product and hosted API.

10. Update an existing clone

```
git status
git pull --ff-only origin main
python -m pip install -r requirements.txt
pnpm install --frozen-lockfile
```

For Docker, use docker compose up --build after pulling changes.

11. Security and data rules

- Never commit .env, credentials, patient documents, local databases, uploads, or logs.
- Use synthetic or properly anonymized data only.
- Rotate a key immediately if it appears in chat, logs, screenshots, or Git history.
- Replace demo JWT login with approved OIDC/OAuth 2.1 before non-demo use.
- The system is administrative; it must not diagnose, prescribe, dose, or interpret findings.

```
git status
git diff --check
git grep -n "OPENAI_API_KEY="
```

12. Troubleshooting

Symptom	Likely check	Resolution
Backend offline	/health on port 8000	Start API; verify API base URL
Port conflict	Local process/container	Stop conflict or update URLs/origins together
Login fails	Seeded database	Run app.seed or recreate local Docker volume
LLM stays fallback	Flag, key, quota, restart	Set both values; use active quota; restart
Upload fails	Size/type/directory	Use synthetic file under 10 MB; check write access
CORS error	Allowed origins/API URL	Use localhost origins and matching API base
No slot	Seed/bookings	Use another specialty/date or fresh local database
Build failure	Node/pnpm/lockfile	Use Node 22.13+ and frozen install
CI syntax mismatch	Python version	Use Python 3.11/3.12; run tests

13. Completion checklist

Check	Pass condition
Repository	main selected; clean; no secrets
Configuration	Ignored local .env files created
Database	Synthetic seed completed
API	/health returns status: ok
Web	Port 3000 opens and patient login works
Roles	Staff/reviewer case access works
Workflow	Normal journey reaches slot confirmation
Human review	Paused case resumes after approval
Documents	Mismatch remains missing and warns
Safety	Clinical request escalates without advice
Tests	Backend, lint, build, and UI tests pass

When every row passes, the clone is ready for development or evaluation.

Repository: https://github.com/chinmayk1613/AgentCare_AI_Patient_Administration_and_Care_Coordination